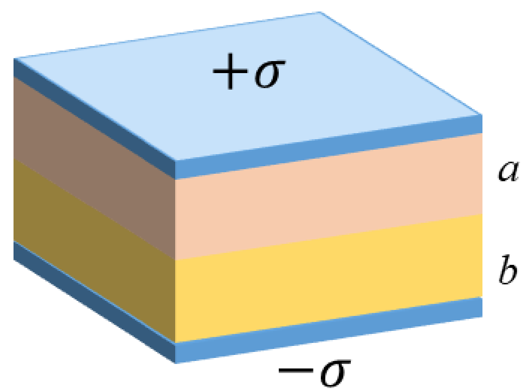
Exercise Session 8
Electric Fields in Matter 3 + Magnetostatics 1

Question 1:

Let's assume σ (Units: $\mathrm{C/m^2}$) is the free surface charge density placed on the metal plates (not in the dielectric). So, the top plate carries $+\sigma$ and the bottom plate carries $-\sigma$.

Two dielectric slabs fill the space between the plates:

- Slab 1 (upper slab): thickness a, relative permittivity $\varepsilon_{r1} = 3$.
- Slab 2 (lower slab): thickness b, relative permittivity $\varepsilon_{r2} = 2$.



The plates are horizontal. Define $+\hat{\mathbf{z}}$ downward, from the top plate to the bottom plate. Then the electric field $\mathbf{E}$ and displacement field $\mathbf{D}$ point in $+\hat{\mathbf{z}}$.

Tasks

(a) Find $\mathbf{D}$ in each slab (include direction) and conceptually answer why is $\mathbf{D}$ the same in both slabs?
(b) Find $\mathbf{E}$ in each slab (include direction) and conceptually answer which slab has the larger $|\mathbf{E}|$, and why?
(c) Find the polarization $\mathbf{P}$ in each slab (include direction).
(d) Find the potential difference V between the plates (top to bottom).
(e) If $\sigma$is doubled, what happens to $\mathbf{D}$, $\mathbf{E}_1$, $\mathbf{E}_2$, and V?

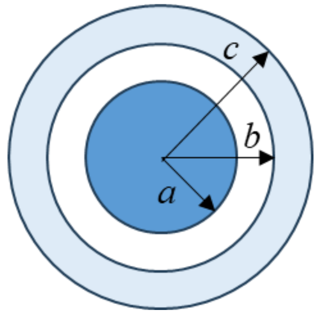
Question 2:

A coaxial cable consists of a solid copper inner conductor of radius a, surrounded by a concentric copper shell whose inner radius is c. The region between the conductors is filled with two dielectric layers:

i) vacuum (or air) for $a < r < b$, and

ii) a dielectric of relative permittivity $\varepsilon_r$for $b < r < c$, as shown in the figure.

The inner conductor carries a free line charge density $+\lambda$(C/m). Since the cable is overall electrically neutral, the outer conductor carries a total line charge density $-\lambda$. Find the capacitance per unit length of the cable.



Question 3:

You have an insulating sphere of radius a with relative permittivity ε_r (inside the sphere). The sphere contains a non-uniform free volume charge density [ρ_f is non-uniform and it increases linearly with r (zero at $r = 0$, largest at $r = a$)]

$$\rho_f(r) = 4kr(0 \leq r \leq a),$$

where k is a constant. Outside the sphere ($r > a$) is vacuum.

(a) Find $\mathbf{D}(r)$ and $\mathbf{E}(r)$ inside and outside the sphere.

(b) Find the electrostatic field energy

$$W_{\text{field}} = \frac{\varepsilon_0}{2}\int E^2\, d\tau,$$

and the total energy (including polarization energy)

$$W_{\text{total}} = \frac{1}{2}\int \mathbf{D} \cdot \mathbf{E}\, d\tau.$$

Assume spherical symmetry; give final answers in terms of $k, a, \varepsilon_0, \varepsilon_r$.

Hint: for a spherical Gaussian surface of radius r, the enclosed free charge is

$$Q_{f\text{enc}}(r) = \int_0^r \rho_f\,(r')\, 4\pi r'^2\, dr'.$$

Question 4:

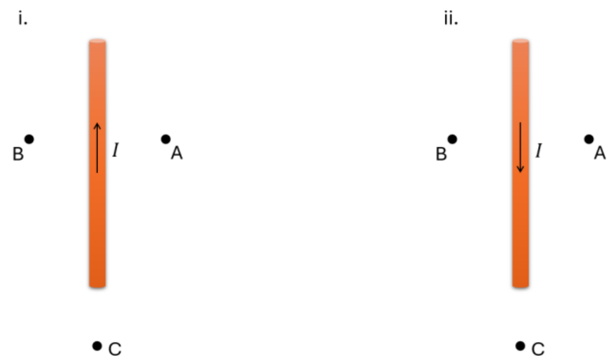
a) Suppose that you are out on a mountain holding a compass. In which direction will the compass needle point and why?

b) Suppose now that you bring the compass close to a wire hanging straight from the ceiling, carrying current. In which direction will the compass needle point when you hold the compass at each of the positions A, B and C in the figures below if

   i. the current flows in the upward direction (see figure to the left)?

   ii. the current flows in the downward direction (see figure to the right)?

You hold the compass flat facing upwards at all positions and look at it from above.



c) Consider now that you have both these wires next to each other, one carrying a current in the upward direction and the other in the downward direction. What will be the direction of the Lorentz force on each wire from the other? Will the wires attract or repel?

<u>Question 5:</u>

(a) A point charge $+q$ is held fixed in space. Can it produce any magnetic field along with its electric fields?
(b) The same charge now moves with constant velocity **v**. Compared to the stationary case, what additional field appears?
(c) Write the **Lorentz force** on a test charge Q moving with velocity **v** in:
(i) a magnetic field only, **B**
(ii) both electric and magnetic fields, **E** and **B**
(d) Let's assume a positively charged particle q enters a uniform magnetic field **B** pointing out of the page and the electric force **E** points to the left ($-\hat{x}$). Find and sketch the trajectory of the particle, if it starts at the origin with velocity:
(i) $\mathbf{v}(0) = (E/B)\,\hat{\mathbf{y}}$,
(ii) $\mathbf{v}(0) = (E/2B)\,\hat{\mathbf{y}}$,

Can the magnetic field accelerate the velocity of the charged particle by making it go faster? Briefly explain.